小野 method to Higgs Mode

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1 Higgs モード関連

1.1 Hamiltonian

$$\mathbf{\Psi_k} = \begin{pmatrix} c_{k\uparrow} \\ c^\dagger_{-k\downarrow} \end{pmatrix}, \tag{1}$$

$$\mathbf{\Psi_k}^\dagger = \begin{pmatrix} c^\dagger_{k\uparrow} & c_{-k\downarrow} \end{pmatrix} \tag{2}$$

$$H = -\sum_{\mathbf{k}} (c^\dagger_{k\uparrow} c_{-k\downarrow} + \text{h.c.}) \ + \ \frac{1}{U}|\Delta|^2 \tag{3}$$

$$= \sum_{\mathbf{k}} \mathbf{\Psi_k}^\dagger \begin{pmatrix} \xi_k & \Delta \\ \Delta* & -\xi \end{pmatrix} \mathbf{\Psi_k} \tag{4}$$

$$= \sum_{\mathbf{k}} \mathbf{\Psi_k}^\dagger H_k \mathbf{\Psi_k} \tag{5}$$

$$\tag{6}$$

We have employed 2D-square lattice

$$\xi_k = -2t(\cos(k_x t) + \cos(k_y t)) \ - \ \mu \tag{7}$$

Peierls substitution: $k \to k - A(t)$

$$H_{k-A(t)} = \begin{pmatrix} \frac{\xi_{k-A(t)} + \xi_{k+A(t)}}{2} & -\Delta(t) \\ -\Delta(t) & -\frac{\xi_{k-A(t)} + \xi_{k+A(t)}}{2} \end{pmatrix} \tag{8}$$

The term

$$\frac{\xi_{k-A(t)} \ - \ \xi_{k+A(t)}}{2} \tau_0 \tag{9}$$

is omitted because it does not evolve in time

1.2 Pseudo spin

The pseudo spin is defined as:

$$\sigma_k^\alpha = \frac{1}{2}\langle \mathbf{\Psi_k}^\dagger(t)\tau_\alpha \mathbf{\Psi_k}(t)\rangle \tag{10}$$

Effective magnet

$$\mathbf{b_k} = \begin{pmatrix} -\mathrm{Re}\Delta(t) & \mathrm{Im}\Delta(t) & \frac{\xi_{k-A(t)}+\xi_{k+A(t)}}{2} \end{pmatrix} \tag{11}$$

Δ is determined by pseudo spin σ as following

$$\Delta = \frac{U}{N}\sum_{\mathbf{k}}(\sigma_{\mathbf{k}}^x + i\sigma_{\mathbf{k}}^y) \tag{12}$$

Therefore, you have to determine Δ, $\mathbf{b}$, σ self-consistently

$$\Delta_0 \to \mathbf{b_0} \to \sigma_\mathbf{0} \to \mathbf{\Delta_1} \to \mathbf{b_1} \to \sigma_\mathbf{1} \to \mathbf{\Delta_2} \to ... \tag{13}$$

Initial pseudo spin

$$\sigma_k^\alpha = \frac{1}{2}\mathrm{Tr}(\rho\tau_\alpha) \tag{14}$$

$$= \frac{1}{2}\mathrm{Tr}(U\boldsymbol{f}U^\dagger\tau_\alpha) \tag{15}$$

$$= \frac{1}{2}\mathrm{Tr}(\boldsymbol{f}U^\dagger\tau_\alpha U) \tag{16}$$

Where, $\boldsymbol{f}$ is

$$\boldsymbol{f} = \begin{pmatrix} f(E_k) & 0 \\ 0 & f(-E_k) \end{pmatrix} \tag{17}$$

$f(\varepsilon)$ is fermi distribution function

Derivaton

$$\frac{\partial}{\partial t}\boldsymbol{\sigma_k} = 2\mathbf{b_k} \times \boldsymbol{\sigma_k} \tag{18}$$

analogy from Bloch equation, the hamiltonian ia described as following

$$H = \sum_{\mathbf{k}} 2\mathbf{b_k}\cdot\boldsymbol{\sigma_k} \tag{19}$$

Time evolution with Sympson's rule

$$\boldsymbol{\sigma_{i+2}} = \boldsymbol{\sigma_i} + \frac{\Delta t}{3}(2\mathbf{b_i} \times \boldsymbol{\sigma_i} + 8\mathbf{b_{i+1}} \times \boldsymbol{\sigma_{i+1}} + 2\mathbf{b_{i+2}} \times \boldsymbol{\sigma_{i+2}}) \tag{20}$$

For σ_2, I used trapzoid rule with sub-mesh of time

$$\boldsymbol{\sigma_{i+1}} = \boldsymbol{\sigma_i} + \frac{\Delta t}{2}(2\mathbf{b_i} \times \boldsymbol{\sigma_i} + 2\mathbf{b_{i+1}} \times \boldsymbol{\sigma_{i+1}}) \tag{21}$$

2 Pulse

In the review paper by Prof.Tsuji, he imposed electric field of

$$E(t) = \begin{cases} E_0 \cos(\Omega t)\sin^2(t/\tau) & (0 \le t < \tau) \\ 0 & (\text{otherwise}) \end{cases} \tag{22}$$

And Therefore, the vector potential is given by

$$A(t) = \begin{cases} E_0 \left[-\frac{\sin(\Omega t)}{2\Omega t} + \frac{\sin[(\Omega+\pi/\tau)t]}{4(\Omega+2\pi/\tau)} + \frac{\sin[(\Omega-\pi/\tau)t]}{4(\Omega-2\pi/\tau)} \right] & (0 \le t < \tau) \\ 0 & (\text{otherwise}) \end{cases} \tag{23}$$

In order to make $A(\tau) = 0$ and smooth connection at $t = \tau$,

$$\Omega\tau = n\pi \ (n \neq \pm 2) \tag{24}$$

is needed.

There are delta function peaks with width of $1/\tau$. You can see why by investigating the Fourier transformation of the the envelope.

$$\int_0^\tau dt \sin^2(\pi t/\tau) e^{i\omega t} = \tau \frac{e^{i\omega\tau} - 1}{i} \left[\frac{1}{2\omega\tau} - \frac{1}{4(\omega\tau + 2\pi)} - \frac{1}{4(\omega\tau - 2\pi)} \right] \tag{25}$$

Now, I set $\Omega = 40.0$ and $n = 4$ and therefore $\tau = \pi/10$.

The width in the frequency space $\Delta\omega$ is given by

$$\Delta\omega = 1/\tau \tag{26}$$

And we also employed a gaussian pulse given by

$$f(t) = \frac{F_0}{2\pi} e^{-t^2/(2\tau_f^2)}, \qquad f(\omega) = \frac{F_0 \tau_f}{\sqrt{2\pi}} e^{-\omega^2 \tau_f^2/2} \tag{27}$$

The structure of spectroscopy must not depend on the pulse imposed to the system. We should check whether on not this holds true later.

3 The "Ono method"

Let the test pulse be as follows:

$$g_{\cos_1}(t) = \frac{F_0}{2\pi} e^{-(t/\tau_g)^2} \cos(\omega_1 t) \, , g_{\sin_1}(t) = \frac{F_0}{2\pi} e^{-(t/\tau_g)^2} \sin(\omega_1 t) \tag{28}$$

the first derivation of $\langle\Delta\rangle^{(2)}$ is

$$\delta_g \langle\Delta\rangle^{(2)} \equiv \langle\Delta\rangle^{(2)}_{f+\epsilon g} - \langle\Delta\rangle^{(2)}_f \tag{29}$$

$$= \epsilon \int_{-\infty}^{\infty} d\omega_1 g(\omega_1) \frac{\chi^{(2)}(\omega_1, \omega - \omega_1) f(\omega - \omega_1)}{2\pi} + O(\epsilon^2) \tag{30}$$

When you impose pulses such f, $f + \varepsilon g_{\cos}$, $f + \varepsilon g_{\sin}$ and set $\tau_g \to \infty$, you will get

$$\chi^{(2)}(\omega_1, \omega - \omega_1) = \frac{2\pi}{f(\omega - \omega_1)} \left[\frac{\delta_{g_{\cos_1}} Q^{(2)}(\omega)}{\varepsilon F_0} - i \frac{\delta_{g_{\sin_1}} Q^{(2)}(\omega)}{\varepsilon F_0} \right] \tag{31}$$

4 Parameters

Time Mesh

$$
\begin{aligned}
N_t &= 7001 \\
t_{\min} &= -700 \\
t_{\max} &= 700 \\
\text{Temperature} &= 0.02 \\
U &= 4.0
\end{aligned}
$$

k-mesh

$$
\begin{aligned}
N_{k_x} \times N_{k_y} &= 50 \times 50 \\
k_{\min} &= 0 \\
k_{\max} &= 2\pi
\end{aligned}
$$

ω-mesh

$$
\begin{aligned}
N_{\omega_1} \times N_{\omega_2} &= 201 \times 201 \\
\omega_{\min} &= 0 \\
\omega_{\max} &= 2\pi
\end{aligned}
$$

Hamiltonian

$$
\begin{aligned}
t &= 1.0 \\
\mu &= 0.0
\end{aligned}
$$

[pulse]

In Eq. (22):

$\Omega = 40$

$\tau = \pi/10$

In the Ono method, two pulses with different amplitudes are imposed $F_0^{(1)}$ and $F_0^{(2)}$

$F_0^{(1)} = 10^{-3}$

$F_0^{(2)} = 10^{-2}$

In Eq. (27):

$\tau_f = 0.1$

$\tau_g = \sqrt{2}/0.01$

$F_0^{(1)} = 10^{-3}$

$F_0^{(2)} = 10^{-2}$

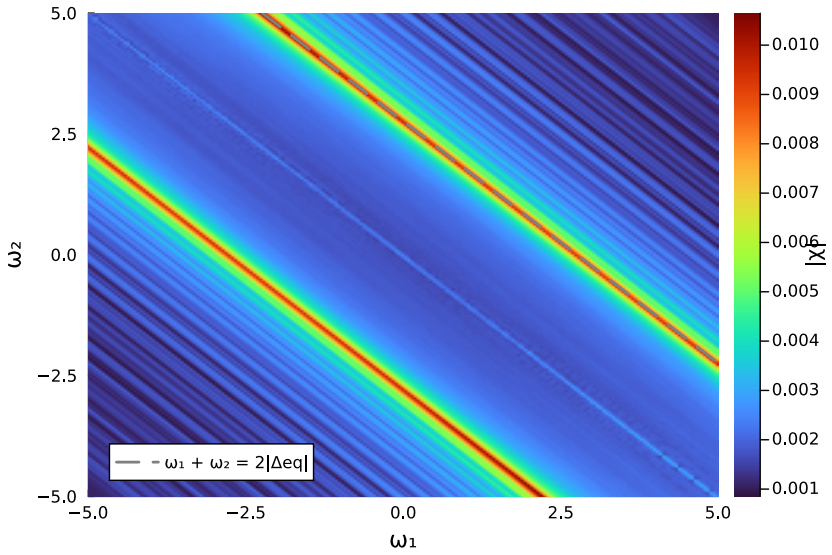
$\varepsilon = 10^{-3}$

5 Graphs

Before showing the graphs, Remember we performed calculation on double precision, even though we should have used double-double precision due to the sensitivity of high-order calculations, which causes unexpected behavior.

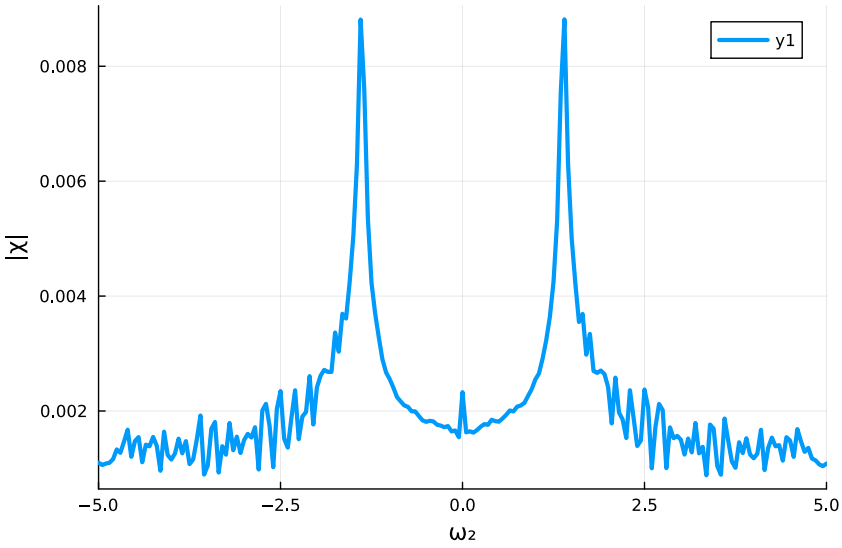
5.1 Eq.(23)

The following result is where the pulse is given in Eq.(23)



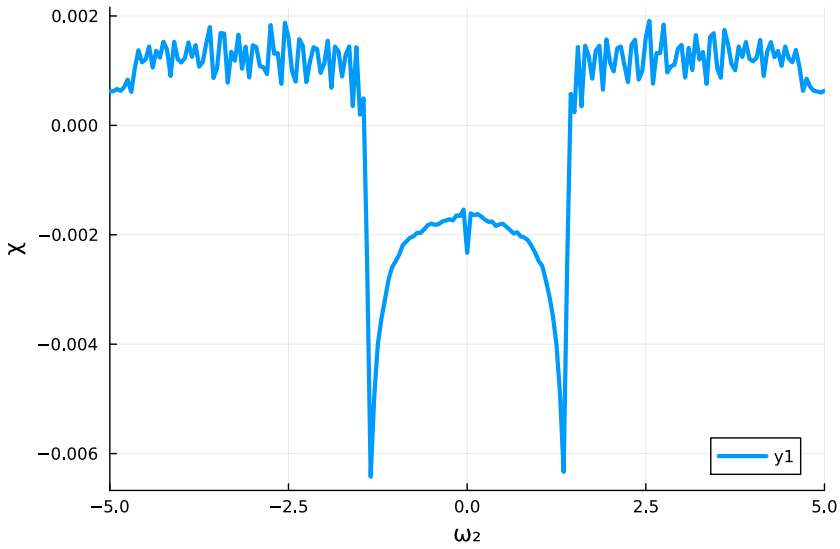
heightheight

図 1　Heatmap of $|\chi^{(2)}(\omega_1, \omega_2)|$



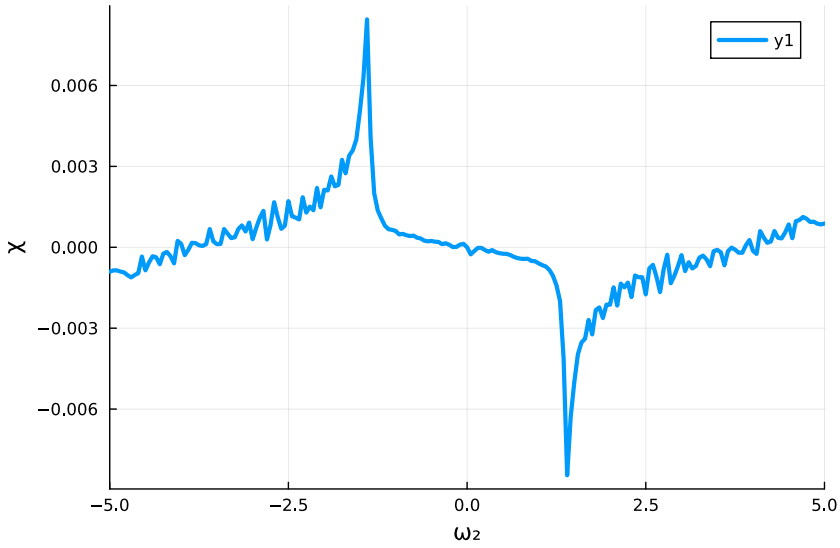
heightheight

図 2　$|\chi^{(2)}(\omega_1, \omega_2)|$ in the line $\omega_1 = \omega_2$



heightheight

図 3　$\mathrm{Re}\chi^{(2)}(\omega_1, \omega_2)$ in the line $\omega_1 = \omega_2$

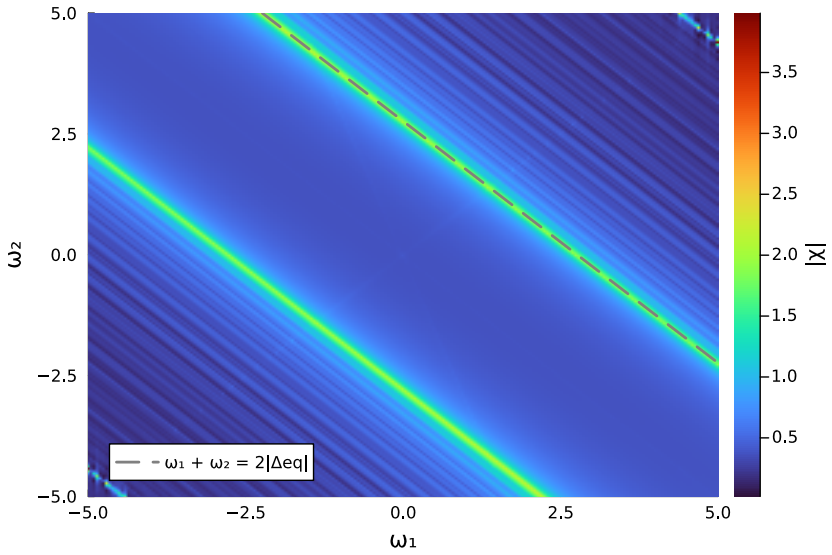


heightheight

図 4　$\mathrm{Im}\chi^{(2)}(\omega_1, \omega_2)$ in the line $\omega_1 = \omega_2$

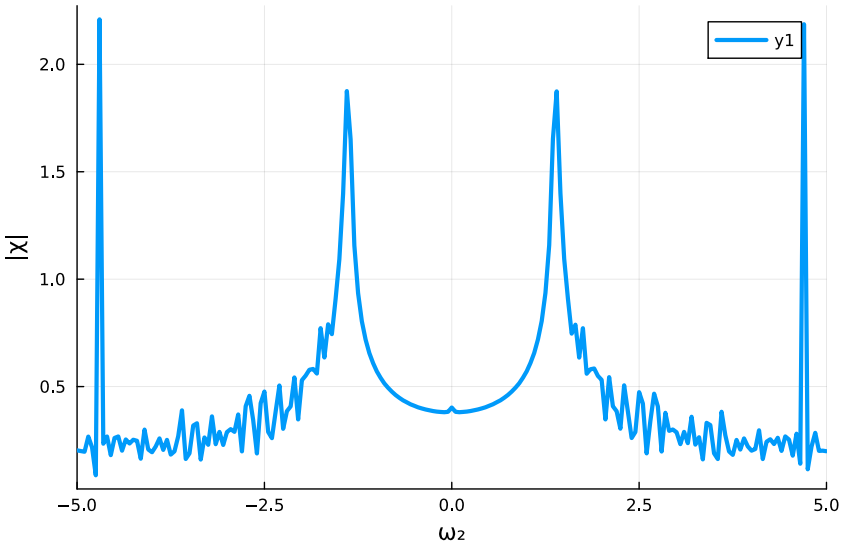
5.2　Eq.(27)

The following graph is where the pulse is given in Eq.(27)



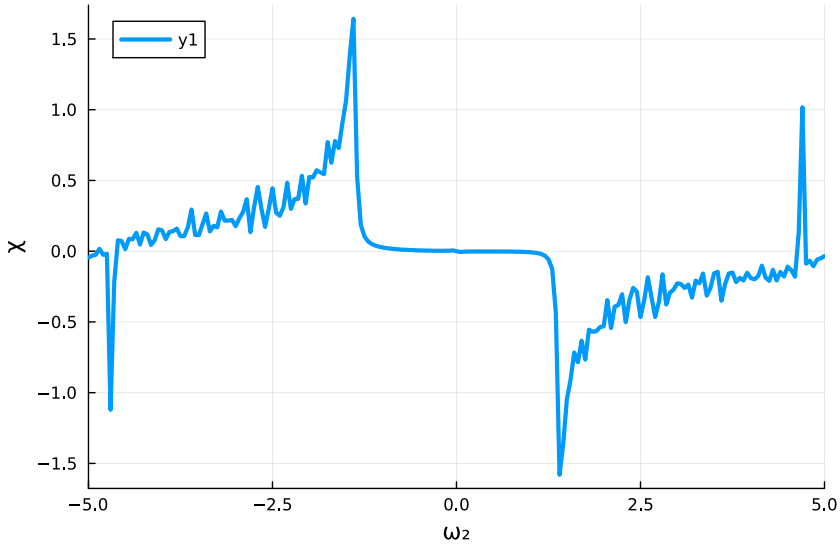
heightheight

図 5　Heatmap of $|\chi^{(2)}(\omega_1, \omega_2)|$



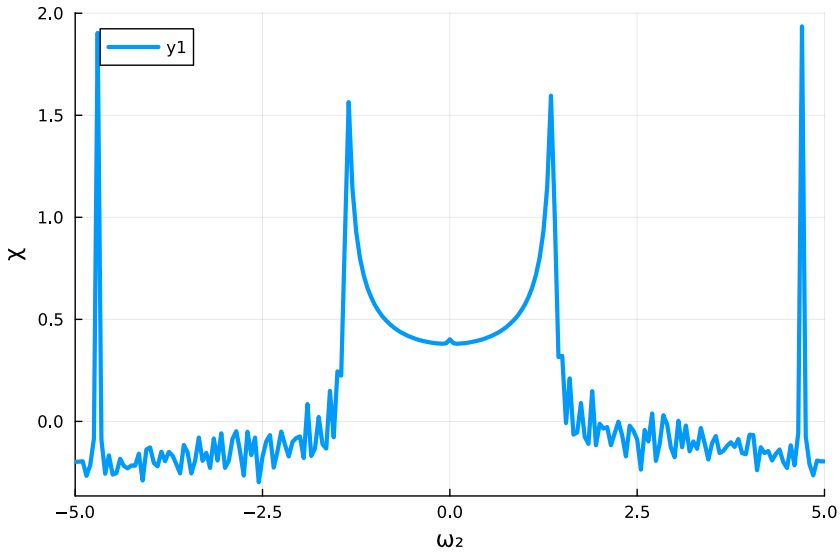
heightheight

図 6　$|\chi^{(2)}(\omega_1, \omega_2)|$ in the line $\omega_1 = \omega_2$



heightheight

図 7　Re$\chi^{(2)}(\omega_1, \omega_2)$ in the line $\omega_1 = \omega_2$



heightheight

図 8　Im$\chi^{(2)}(\omega_1, \omega_2)$ in the line $\omega_1 = \omega_2$

It seems that both graphs look similar except the peak at $\omega_2 \approx \pm 5.0$ and I do not know what makes this difference.